

ITERATION 0

Starting: $m = [-1, 2]$ $b = [1, 1]$ $x = [1, 3]$ $y = 5$
 $x = [4, 10]$ $y = 6$

Predictions: $\hat{y}_1 = m_1 x_1 + m_2 x_2 + b_1$
 $= -1 \times 1 + 2 \times 3 + 1$
 $= -1 \times 6 + 1$
 $\hat{y}_1 = -5$

$\hat{y}_2 = -1 \times 4 + 2 \times 10 + 1$
 $= -4 \times 20 + 1$
 $\hat{y}_2 = -79$

Compute Errors ($e = \hat{y} - y$)

$e_1 = \hat{y}_1 - y_1 = -5 - 5 = -10$

$e_2 = \hat{y}_2 - y_2 = -79 - 6 = -85$

Compute Gradient: $\frac{dJ}{dm_1} = m_2 (3e_1 + 40e_2)$
 $= 2 (3 \times -10 + 40 \times -85)$

$dJ/dm_1 = -6860$

$\frac{dJ}{dm_2} = m_1 (3e_1 + 40e_2)$
 $= -1 (3 \times -10 + 40 \times -85)$

$dJ/dm_2 = 3430$

$\frac{dJ}{db_1} = e_1 = -10$

$\frac{dJ}{db_2} = e_2 = -85$

Update Parameters: $\alpha = 0.00001$ (Because gradient magnitude is large)

New parameter = Old parameter - $\alpha \frac{dJ}{d\text{parameter}}$

$$\text{For } m_1 = -1 - 0.00001(-6860) \\ = -0.9314$$

$$m_2 = 2 - 0.00001(3430) \\ = 1.9657$$

$$b_1 = 1 - 0.00001(-10) \\ = 1.0001$$

$$b_2 = 1 - 0.00001(-85) \\ = 1.00085$$

Updated state after iteration 0:

$$m = [-0.9314, 1.9657]$$

$$b = [1.0001, 1.00085]$$